Multivariate Calculus Lecture Notes
Section 12.1 – Vectors in the Plane

I. Basics
  a. A *vector* is an entity consisting of both length and direction.
    i. A *scalar* merely has length.
    ii. For example, velocity is a vector, while speed is a scalar.
  b. Representations (do graphical examples of each)
    i. Directed line segments
    ii. Length and angle
    iii. Component form
      1. The initial point is at the origin.
      2. The terminal point is $\langle v_1, v_2 \rangle$.
      3. The values of v_1 and v_2 are the components of $\mathbf{v}$.
    iv. Two vectors with the same length and direction are said to be *equivalent*.
  c. Zero vector
    i. It has no length, so the concept of direction is meaningless.
    ii. It is often represented as $\mathbf{0}$.
  d. Magnitude of a vector
    i. How do you find the length of a vector $\mathbf{v}$ with origin at $(3, 4)$ and terminus at $(1, 5)$?
      1. First, write the vector in its component form: $\mathbf{v} = \langle -2, 1 \rangle$.
      2. Then use the distance formula: $\|\mathbf{v}\| = \sqrt{4 + 1} = \sqrt{5}$.
    ii. How do you find the angle the vector makes with the positive horizontal axis?
      1. Graphically:

      2. We have $\sin\theta = 1/\sqrt{5}$.
      3. So $\theta = \arcsin(1/\sqrt{5})$.
        - You get approximately 0.4636 radians.
        - However, this angle is in the first quadrant, and the angle we want is in the second quadrant.
        - The angle we want is $\pi - 0.4636 \approx 2.678$ radians.
        - Remember, the arcsine function returns values between $-\pi/2$ and $\pi/2$.
    iii. The length of a vector is sometimes called its *magnitude* or its *norm*.
  e. You can also have vectors in three, four, or however many dimensions you want.
II. Arithmetic operations
  a. Addition
    i. To add two vectors you simply add their components.
    ii. For example, $\langle -2, 1 \rangle + \langle 1, 3 \rangle = \langle -1, 4 \rangle$.
    iii. Geometrically:

b. Scalar multiplication
   i. To multiply a scalar by a vector you multiply each component by the scalar.
   ii. For example, $3\langle -2, 1\rangle = \langle -6, 3\rangle$.
   iii. Geometrically:

c. Negation
   i. The same principle works when you multiply by -1: $-1\langle -2, 1\rangle = \langle 2, -1\rangle$.
   ii. Geometrically:

d. Subtraction
   i. To visualize the subtraction of two vectors geometrically, think of adding the negative of the second to the first.
   ii. For example, $\langle -2, 1\rangle - \langle 1, 3\rangle = \langle -2, 1\rangle + \langle -1, -3\rangle = \langle -3, -2\rangle$.
   iii. Geometrically, $\mathbf{u} - \mathbf{v}$ is the vector from the end of $\mathbf{v}$ to the end of $\mathbf{u}$.

e. Vector operations satisfy many of the arithmetic properties that scalars satisfy.
   i. For example, addition is associative and commutative, scalar multiplication distributes over addition, $\mathbf{0}$ is the identity vector, each vector has an additive inverse.
   ii. See page 628 in the text.

f. Length of a scalar multiple
   i. Example: $||3\mathbf{v}||$, where $\mathbf{v} = \langle -2, 1\rangle$.
      1. $3\mathbf{v} = \langle -6, 3\rangle$.
      2. So the magnitude is $\sqrt{36 + 9} = \sqrt{45} = 3\sqrt{5}$.
   ii. In general, we have $||k\mathbf{v}|| = |k|\,||\mathbf{v}||$.
      1. $||k\mathbf{v}|| = ||\langle kv_1, kv_2\rangle|| = \sqrt{k^2v_1^2 + k^2v_2^2} = \sqrt{k^2(v_1^2 + v_2^2)} = |k|\sqrt{v_1^2 + v_2^2} = |k|\,||\mathbf{v}||$.
      2. This proof is for two-dimensional vectors, but the obvious extensions work in higher dimensions.

III. Parallel vectors
   a. Two vectors $\mathbf{u}$ and $\mathbf{v}$ are *parallel* if $\mathbf{u} = c\mathbf{v}$ for some constant c.
   b. For example, $\langle -2, 1\rangle$ and $\langle 1, -1/2\rangle$ are parallel.

IV. Unit vectors
   a. A *unit vector* is a vector of length 1.
   b. Unit vectors tend to show up in applications.
   c. Standard unit vectors
      i. The *standard unit vectors* in two dimensions are $\mathbf{i} = \langle 1, 0\rangle$, $\mathbf{j} = \langle 0, 1\rangle$.

ii. Any vector $\langle v_1, v_2 \rangle$ can be expressed as a linear combination of the standard unit vectors, like so: $v_1\mathbf{i} + v_2\mathbf{j}$.

d. How do you obtain a unit vector in the direction of some given nonzero vector $\mathbf{v}$?
   i. To retain the direction but change the magnitude you need to multiply by some scalar.
   ii. Multiply $\mathbf{v}$ by $1/||\mathbf{v}||$.
   $$\left\| \frac{1}{||\mathbf{v}||}\mathbf{v} \right\| = \frac{1}{||\mathbf{v}||}\|\mathbf{v}\| = 1$$
   iii. Example
      1. A unit vector in the direction of $\mathbf{v} = \langle -2, 1 \rangle$ is $\langle -2/\sqrt{5}, 1/\sqrt{5} \rangle$, since $\|\mathbf{v}\| = \sqrt{5}$.
      2. To verify, the magnitude of the second vector is $4/5 + 1/5 = 1$, as it should be.
      3. So, if $\mathbf{v} = \langle -2, 1 \rangle$, then we could also write $\mathbf{v} = \sqrt{5}\langle -2/\sqrt{5}, 1/\sqrt{5} \rangle = \|\mathbf{v}\|\mathbf{e_v}$, where $\mathbf{e_v}$ is a unit vector in the direction of $\mathbf{v}$.
      4. Since $\mathbf{e_v}$ is a unit vector, we could also write $\mathbf{e_v} = \langle \cos\theta, \sin\theta \rangle$, where $\theta \approx 2.678$ radians and is the angle that $\mathbf{v}$ makes with the positive horizontal axis.

V. Triangle inequality

a. It's pretty clear that for any triangle, the lengths of the two sides added together is bigger than the length of the third.
   i. Graphically:

   ii. In fact, it's just an application of the principle that the shortest distance between two points is a straight line.

b. If we represent this idea in terms of vectors, we get what's known as the *triangle inequality*.
   i. For any two vectors $\mathbf{v}$ and $\mathbf{w}$,
   $$||\mathbf{v} + \mathbf{w}|| \leq ||\mathbf{v}|| + ||\mathbf{w}||.$$
   ii. Graphically:

c. In order to have equality, we must have one of the following.
   i. $\mathbf{v} = \mathbf{0}$
   ii. $\mathbf{w} = \mathbf{0}$
   iii. $\mathbf{w} = \lambda\mathbf{v}$, where $\lambda \geq 0$.

Homework: Section 12.1 #10, 14, 21, 25, 28, 30, 43, 47, 48, 62, 65.